

VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-Language Models (VLMs) are compelling candidates for zero-shot goal-conditioned value functions, yet their frozen pre-trained representations fundamentally limit generalization and temporal reasoning. We introduce VITA, a zero-shot value function learning method that overcomes both limitations through test-time adaptation. At inference, a lightweight adaptation module is updated with a single gradient step on a meta-learned self-supervised loss, so that every test-time update sharpens value estimation. By updating sequentially over a trajectory, VITA encodes history directly into its parameters, resolving the temporal reasoning limitations that constrain prior work. To eliminate shortcut learning, we introduce a dissimilarity-based sampling strategy that selects semantically diverse segments of the trajectory during training. On real-world robotic manipulation tasks, VITA generalizes from a single training environment to diverse out-of-distribution tasks, environments, and embodiments, consistently outperforming the state-of-the-art zero-shot method built on autoregressive VLMs. We further show that VITA’s zero-shot value estimates provide effective reward shaping for offline reinforcement learning, yielding multi-task policies on the Meta-World benchmark that surpass those trained with the simulation’s hand-designed fuzzy-logic dense rewards.

1 INTRODUCTION

Vision-Language Models (VLMs) achieve strong generalization across diverse tasks and domains by learning from large-scale, unstructured web data without human supervision (Radford et al., 2021; Alayrac et al., 2022). Robotics and embodied AI have yet to reach comparable generality: despite significant advances in learning generalist policies for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024), state-of-the-art methods remain bottlenecked by their reliance on expert demonstrations (Ho & Ermon, 2016). World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) offer a compelling path forward, enabling agents to learn in simulated environments, with applications spanning robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). Yet even as world models generate realistic visual trajectories in diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), one challenge remains decisive: *how can agents effectively learn from videos at scale without relying on human supervision?*

One line of research learns policies directly from expert visual trajectories by inferring actions from visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). These latent actions can be mapped to executable controls at deployment, though doing so reliably remains difficult in practice. A complementary and, we argue, more scalable direction learns a universal *goal-conditioned value function* and uses it as a zero-shot reward shaping and supervision signal for reinforcement learning (RL) and imitation learning, respectively (Chen et al., 2021; Ma et al., 2022). In this framework, goal-conditioned value estimation reduces to a clear objective: predicting how far an agent has progressed toward completing a task from visual observations and a natural language task description (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025).

Prior work has employed pre-trained contrastive VLMs (Radford et al., 2021) for zero-shot reward shaping, leveraging the similarity between task descriptions and visual observations in a shared multimodal representation space to ground progress in semantic context (Ma et al., 2023b; Baumli

et al., 2023; Rocamonde et al., 2024). These methods, however, cannot capture the temporal context needed to disambiguate visually similar states that occur at different stages of a task (e.g., folding vs. unfolding a shirt). Autoregressive VLMs (Alayrac et al., 2022) do incorporate temporal context by conditioning on the entire visual trajectory within the prompt, but they inherit a strong bias toward monotonically increasing predictions from the chronologically ordered datasets used in pre-training (Ma et al., 2024). Crucially, both architectures depend on *frozen* pre-trained representations (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023) for zero-shot prediction, and this dependence is precisely what caps their generalization and temporal reasoning (Pătrăucean et al., 2023). Existing remedies—large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), and demo-based reward specification (Sontakke et al., 2023)—all buy generalization at the cost of additional data, supervision, or compute.

We introduce VITA, a zero-shot value function learning method that directly attacks both limitations—generalization and temporal reasoning—through test-time adaptation, and outperforms the state-of-the-art zero-shot approach (Ma et al., 2024) on real-world robotic manipulation tasks. Rather than relying on large-scale pre-training or expert demonstrations for generalization, VITA adapts online to both the semantic and temporal context of each trajectory. At inference, a lightweight adaptation module is updated in negligible time with a single gradient step on a meta-learned self-supervised loss (Sun et al., 2024), so that every test-time update measurably improves value function estimation (Finn et al., 2017). By updating sequentially over a test trajectory, the value function estimator encodes trajectory history directly into its parameters (Sun et al., 2024), resolving the temporal reasoning limitations of prior work. To eliminate shortcut learning (Geirhos et al., 2020), we introduce a dissimilarity-based sampling strategy that forces reliance on semantic cues, and we validate its effect empirically. Across the core capabilities of value function estimation—generalization under distribution shifts, discrimination between expert and non-expert trajectories, and reward shaping for offline RL (Levine et al., 2020)—our evaluation establishes the effectiveness of VITA.

We summarize our key contributions as follows:

- We propose VITA, the first test-time adaptation method that equips contrastive VLMs with both the generalization and the temporal reasoning required for zero-shot value function estimation—without any task-specific demonstrations or large-scale pre-training.
- We show that VITA generalizes from a single training environment to diverse out-of-distribution tasks, environments, and embodiments in robotic manipulation, consistently outperforming the state-of-the-art zero-shot method (Ma et al., 2024).
- We demonstrate that VITA’s zero-shot value estimates yield offline RL policies on the Meta-World multi-task benchmark (MT10) that surpass those trained with the simulation’s hand-designed fuzzy-logic dense rewards (Yu et al., 2020).

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

In video trajectories, task progress estimation is naturally cast as goal-conditioned value function estimation, where the reward reflects task completion. We therefore formulate learning a goal-conditioned value function as predicting the degree of task completion from vision–language representations. Formally, the value function is $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$, mapping an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar indicating predicted progress toward goal completion. We set $V(o_t; g) = 0$ at the start of the task and $V(o_t; g) = 1$ at completion. Task progress is aligned with temporal position in expert demonstrations, following the standard assumption that such trajectories exhibit monotonically increasing progress toward goal completion (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, temporal progress is defined by normalized timestep indices: $V^{\pi_E}(o_t; g) = \frac{t}{T}$ where T is the trajectory length. Temporal progress thus provides direct supervision for learning a goal-conditioned value function V .

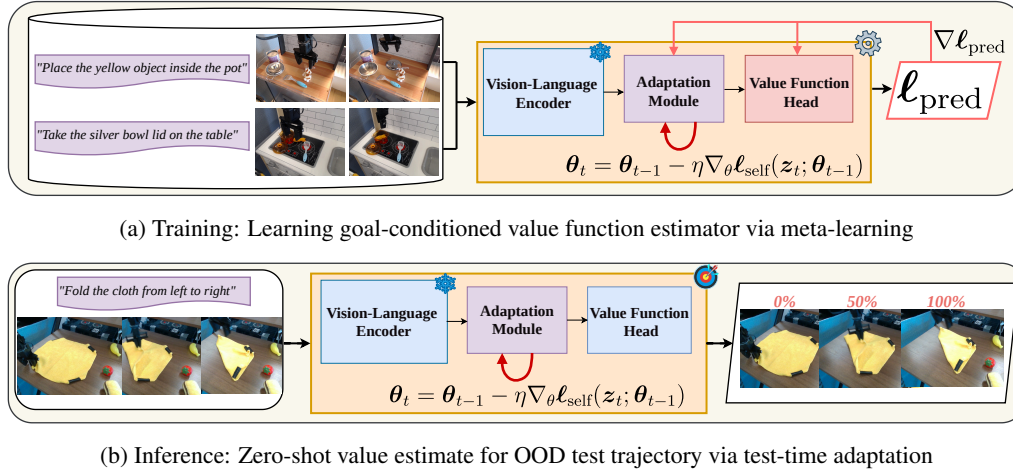


Figure 1: **Overview of VITA.** VITA learns a goal-conditioned value function through meta-learning and generalizes zero-shot to out-of-distribution trajectories through test-time adaptation.

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) treats inference itself as a self-supervised learning task (Goodfellow et al., 2016), updating model parameters on each test instance without access to labels. Originally proposed for static image classification (Sun et al., 2020), TTT has since been extended to sequence modeling (Sun et al., 2024; Wang et al., 2023), where a parametric adaptation module f_{adapt} is updated at each timestep by a gradient step on a self-supervised loss. Unlike standard sequence architectures that encode temporal history in activation states—hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key-value pairs (Vaswani et al., 2017)—TTT encodes history into the *parameters* of f_{adapt} through sequential updates at inference. These parameters act as an implicit memory that encodes temporal history while preserving temporal order. Sun et al. (2024) further meta-learn the self-supervised task under the gradient-based meta-learning paradigm (Finn et al., 2017). In this setting (Sun et al., 2024), the self-supervised loss is parameterized by learnable linear projections and trained to minimize downstream prediction loss *after* a test-time update, rather than being minimized directly. We are the first to bring TTT to goal-conditioned value function learning, capturing temporal context through sequential updates and meta-learning the self-supervised task to improve value estimation instead of fixing it a priori.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

Our goal-conditioned value function estimator comprises three modules: (1) a multimodal encoder that extracts joint visual-language representations from visual trajectories and their task descriptions; (2) an adaptation module updated at test time via a self-supervised loss that is meta-learned to improve value estimation; and (3) a regression head that predicts value estimates. Together, these modules turn a frozen contrastive encoder into a value function that adapts to each trajectory it sees.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We use a frozen contrastive vision-language encoder, CLIP (Radford et al., 2021), to extract representations from visual observations and goal descriptions. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we concatenate their representations at each timestep t into a sequence of joint multimodal representations $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, and ϕ_v and ϕ_g denote the visual and language encoders, respectively. Because CLIP is pre-trained with a contrastive objective that aligns paired image-text inputs in a shared representation space (Radford et al., 2021), representations of visual observations that are semantically closer to goal completion lie closer to the goal description—a property VITA exploits directly.

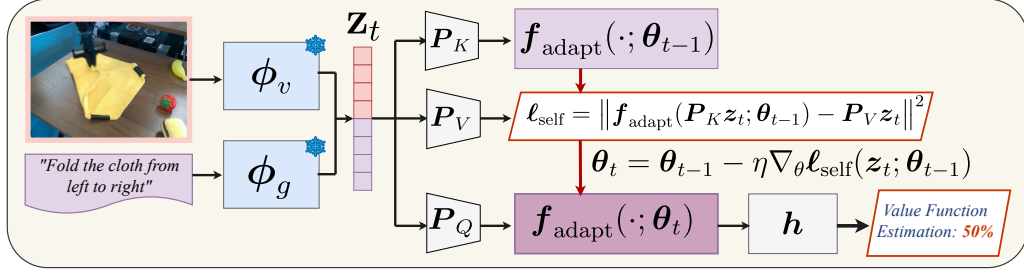


Figure 2: **Test-time Adaptation.** At inference, at each timestep t , the adaptation module f_{adapt} is updated by a gradient step on a meta-learned self-supervised loss ℓ_{self} , encoding temporal history directly into its parameters.

3.1.2 TEST-TIME ADAPTATION

To adapt multimodal representations to both semantic and temporal context, we employ an adaptation module f_{adapt} following the test-time training paradigm of Section 2.2. At inference, the parameters of f_{adapt} are updated online via a gradient step on a meta-learned self-supervised loss ℓ_{self} . We formulate this loss as a reconstruction objective using learnable linear projections P_K and P_V , meta-learned during training so that adaptation improves value estimation. Specifically, with $P_K \in \mathbb{R}^{d' \times 2d}$ generating a perturbed input view and $P_V \in \mathbb{R}^{d' \times 2d}$ producing the target, the self-supervised loss is:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before test-time adaptation. At each timestep t , the parameters are updated by

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . Through these sequential updates, f_{adapt} retains information from past visual observations and thereby captures temporal context. In Section 4.6.2, we show empirically that incremental, per-timestep parameter updates are decisively more effective than updates performed at the trajectory level.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After test-time adaptation, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the input z_t into an adaptation space $\mathbb{R}^{d'}$. Then, this representation is passed through the adaptation module f_{adapt} , followed by a regression head h that outputs predicted values of the goal-conditioned value function:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), mirroring value-function estimator architectures used in deep reinforcement learning (Espeholt et al., 2018). It is trained on the normalized progress labels y_t from expert demonstrations defined in Section 2.1, using a supervised prediction loss ℓ_{pred} equal to the mean squared error between predicted values $V(z_t; g)$ and targets y_t . At inference, we freeze h and update only the adaptation module f_{adapt} online through the self-supervised objective. This design is what allows the value function estimator to generalize zero-shot, with no task-specific demonstrations.

3.2 TRAINING PROCEDURE

We train the value function estimator with gradient-based meta-learning, optimizing the self-supervised loss ℓ_{self} so that test-time adaptation directly improves the supervised prediction loss ℓ_{pred} (Sun et al., 2024). Training operates on diverse sub-trajectories selected by dissimilarity-based sampling, which suppresses shortcut learning and enforces reliance on semantic and temporal cues.

3.2.1 META-LEARNING

During training, the adaptation module f_{adapt} is updated online at each timestep using the self-supervised loss ℓ_{self} . Following the gradient-based meta-learning paradigm (Finn et al., 2017), we differentiate through these adaptation updates with respect to the supervised prediction loss ℓ_{pred} , jointly optimizing the initialization θ_0 of f_{adapt} , the linear projections P_K, P_V, P_Q , and the regression head h . The total training loss combines ℓ_{pred} with ℓ_{self} , weighted by a scalar λ , and the learned θ_0 becomes the initialization for adaptation at inference. In effect, the estimator learns *how to adapt* its internal representations at test time so that value estimation improves.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories contain highly redundant consecutive frames, as observed in prior work on video action recognition (Wang et al., 2018). This redundancy drives the value function estimator toward shortcut cues (Geirhos et al., 2020; Lee et al., 2021b), overfitting to frequently occurring late-stage visual patterns. We eliminate this failure mode with a dissimilarity-based sampling strategy that builds each mini-batch from the most visually diverse sub-trajectories within a trajectory. Doing so maximizes intra-batch variance and acts as importance sampling that explicitly favors semantic diversity.

Formally, given a sequence of multimodal representations $\{z_1, \dots, z_T\}$, we extract fixed-length sub-trajectories using a sliding window of size w_{tr} and stride s , yielding a candidate set $\mathcal{W}$. A diverse subset of size k maximizes pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i is a sub-trajectory of length w_{tr} . Solving Eq. 4 is intractable, requiring enumeration of all $\binom{|\mathcal{W}|}{k}$ subsets. To approximate this objective efficiently, we adopt a scoring-based heuristic, assigning each window $w \in \mathcal{W}$ a score equal to its total dissimilarity with all others:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

By computing a diversity score s for each window and selecting the k highest-scoring windows, we obtain a diverse subset in polynomial time. The overhead is negligible relative to model training, as our complexity analysis in Appendix G confirms. Our ablation study (Section 4.6.1) empirically validates dissimilarity-based sampling against full-trajectory sampling, demonstrating a clear improvement in distinguishing expert from non-expert visual trajectories.

4 EXPERIMENTS

Our evaluation establishes the effectiveness of VITA across the core capabilities of goal-conditioned value functions: (i) generalization in value estimation for real-world robotic manipulation under distribution shifts in task, environment, and embodiment; (ii) discrimination between expert and non-expert visual trajectories in real-world robotic manipulation tasks; (iii) effective zero-shot reward shaping for offline RL in Meta-World MT10, a simulated benchmark for multi-task robot learning.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural language task descriptions from the BridgeData V2 dataset (Walke et al., 2023), which spans a wide range of manipulation tasks, environments, and robot embodiments. We deliberately restrict training to a narrow curated subset of 2,986 expert demonstrations covering only pick-and-place manipulation—excluding folding, sweeping, and stacking tasks entirely. All demonstrations use a single robot embodiment (WidowX 250) across 4 configurations of the ToyKitchen environment. We hold out an additional 287 expert demonstrations as the in-distribution test set, `tk_pnp`. This deliberately narrow training regime lets us test generalization stringently, since every other task, environment, and embodiment in our evaluation is out of distribution. Appendix A includes examples from the training set. We use OpenCLIP ViT-B/32 as the frozen backbone for encoding video frames and task descriptions. The

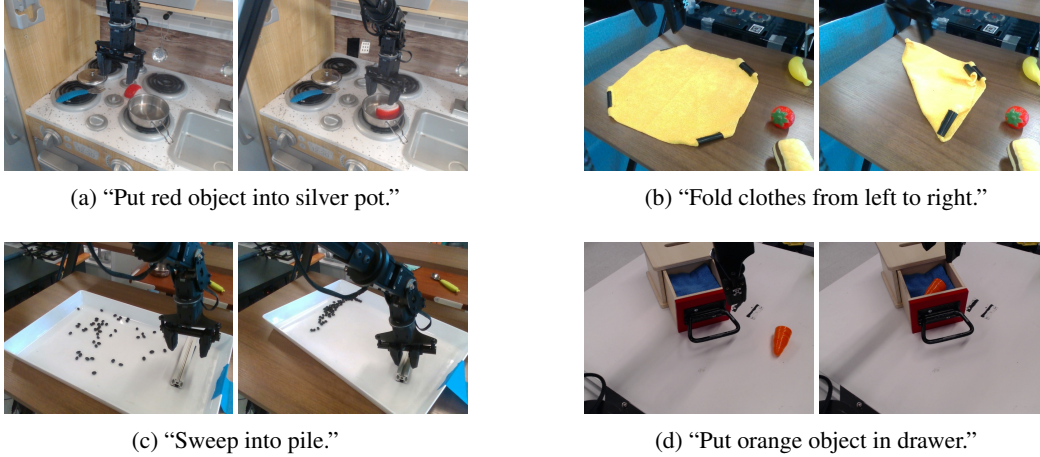


Figure 3: Representative visual trajectories paired with task descriptions spanning the distribution shifts VITA must handle. (a) In-distribution. (b, c) Environment shift. (d) Embodiment and environment shift.

training objective combines ℓ_{pred} with ℓ_{self} , weighted by $\lambda_{\text{self}} = 0.5$. During training, we apply dissimilarity-based sampling with window size $w_{\text{tr}} = 8$, number of sub-trajectories $k = 8$, and stride $s = 1$. At test time, a single gradient step ($t_{\text{ep}} = 1$) using λ_{self} with learning rate $\eta = 0.1$ suffices. Full details and hyperparameter sweeps are reported in Appendix D.

4.2 BASELINES

We benchmark VITA against value functions spanning both contrastive and autoregressive VLMs. VLM-CL computes zero-shot value estimates from cosine similarity between CLIP representations of each frame and the task description (Baumli et al., 2023). VLM-RM regularizes CLIP embeddings by projecting features along the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head on frozen CLIP multimodal representations, and CLIP-GRU trains a Gated Recurrent Unit (GRU) (Cho et al., 2014) on the same representations. GVL (Ma et al., 2024) is the state-of-the-art zero-shot value function built on autoregressive VLMs; GVL-0S denotes the zero-shot setting, where the VLM is prompted with only the test trajectory and task description, and GVL-1S denotes a one-shot in-context setting, where a full shuffled trajectory from the training distribution, together with its progress labels, is supplied as an example. Following GVL, we use Gemini 1.5 Pro (gemini-1.5 Pro-latest) (Gemini Team et al., 2024) with its proposed prompt template. We also tested Qwen-VL 2.5 (Yang et al., 2024), which failed to overcome temporal bias, and GPT-4o (Hurst et al., 2024), which refused to produce value estimates. Additional implementation details appear in Appendix D.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

We test whether progress estimators generalize across novel environments, tasks, and robot embodiments, using subsets of BridgeData V2 curated to introduce variation along each axis. Environment shifts alter scene layout or background: `lm_pnp` is a pick-and-place task in front of a laundry machine, `td_fold`, `ft_fold`, and `rd_fold` feature cloth folding on different surfaces, and `ms_sweep` introduces a long-horizon sweeping task in a confined tray. Embodiment shifts use the DeepThought robot, which differs from the training embodiment (WidowX 250) in both morphology and camera perspective: `dt_tk_pnp` (pick-and-place) and `dt_tk_stack` (stacking) keep the in-distribution environment with a new embodiment, while `dt_ft_stack` (stacking) and `dt_rd_pnp` (drawer pick-and-place) combine embodiment and environment shifts. Each dataset contains 200 expert trajectories with task descriptions, except `ms_sweep`, which contains 100 owing to its size. Appendix B describes every evaluation dataset in full. We measure value function estimation with the *Value Order Correlation* (VOC) metric (Ma et al., 2024), which quantifies the alignment between predicted progress toward task completion and the chronological order of the frames in a visual trajectory.

Table 1: VOC scores for value function estimation under distribution shifts. VITA attains the best or second-best score on every dataset. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Both Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
ES	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

Formally, letting $p_1, \dots, p_T$ denote the predicted progress values for a trajectory of T frames and $r_t = t$ the temporal index, VOC is the Spearman rank correlation ρ_s between the predicted values and the temporal indices.

Table 1 shows that both VITA and CLIP-GRU decisively outperform GVL-0S and GVL-1S, exposing the limitations of pre-trained autoregressive VLMs and confirming that preserving temporal order is essential for effective progress estimation. VITA is the strongest method overall, beating CLIP-GRU on the majority of evaluation datasets (6 out of 10). Both VITA and CLIP-GRU clearly surpass the CLIP-based baselines that lack temporal modeling (CLIP-FT, VLM-CL, VLM-RM), establishing the impact of encoding temporal history on progress estimation. CLIP-FT matches GVL on the in-distribution task but collapses out of distribution, while VLM-CL and VLM-RM perform poorly throughout—a direct consequence of their lack of temporal modeling. For GVL-0S and GVL-1S, in-context examples help in some cases but provide no benefit or even degrade performance in others. Both GVL methods handle folding tasks (td_fold, ft_fold, rd_fold) well yet fail on stacking (dt_tk_stack, dt_ft_stack) and pick-and-place (dt_tk_pnp, lm_pnp) tasks, revealing that the autoregressive VLM underlying GVL is biased toward folding-like manipulations. VITA, by contrast, delivers consistent performance across every task type and distribution shift, demonstrating markedly stronger generalization than in-context learning.

On the long-horizon task ms_sweep, every method except CLIP-FT degrades, yet VITA still achieves the highest VOC score. Its margin over CLIP-GRU here shows that encoding temporal history through test-time adaptation is more effective than recurrent hidden states for long-horizon robotic manipulation. VITA generalizes strongly under embodiment shifts as well. On dt_tk_pnp, which uses a different robot embodiment while sharing the training environment and tasks, both VITA and CLIP-GRU exceed their in-distribution performance, confirming that learned temporal context transfers across embodiments. GVL-1S, in contrast, performs poorly on both dt_tk_pnp and dt_tk_stack, showing that few-shot in-context learning fails to generalize under embodiment shifts in our setup.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond generalization, we probe VITA’s robustness through a demanding test: distinguishing expert from suboptimal visual trajectories by assigning lower progress scores to the latter. We compare model predictions on expert and scripted (non-expert) visual trajectories drawn from the same in-distribution setting of BridgeData V2. The scripted trajectories are generated in the ToyKitchen environment by a heavily randomized controller (Walke et al., 2023), under the same embodiment and environmental configuration as the expert demonstrations. They comprise pick-and-place tasks whose object categories may overlap with the training set: general objects (sc_pnp_obj), rigid items (sc_pnp_robj), soft toys (sc_pnp_stoy), utensils (sc_pnp_uten), and a cloth (sc_pnp_cloth). Although these trajectories match the training distribution in task, embodiment, and environment, their behavior departs from the smooth, monotonic progress of expert demonstrations. Appendix C includes examples.

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-0S	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Success in distinguishing expert from scripted robot demonstrations, measured by average BinVOC across 5 in-distribution scripted datasets; VITA attains perfect discrimination. (b) Offline RL performance on the Meta-World MT10 benchmark, measured by interquartile mean (IQM) with 95% stratified bootstrap CIs; VITA leads all methods, including the simulator’s own dense reward.

Because the scripted datasets come from a randomized controller, they do not reflect quantifiable levels of suboptimality; nonetheless, any effective value function estimator must assign lower VOC scores to suboptimal trajectories than to expert ones in the same setting. We measure this with *BinVOC*, a binary metric defined as $\mathbf{1}[\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, equal to 1 when expert trajectories achieve higher VOC than suboptimal ones and 0 otherwise.

We report discriminative performance as BINVOC averaged over all scripted datasets, measuring expert performance (VOC_{exp}) on the in-distribution test set `tk_pnp`, whose pick-and-place tasks resemble those in the scripted evaluation. VITA, GVL-0S, and GVL-1S achieve perfect discrimination, assigning higher VOC scores to expert trajectories on every task. Notably, VITA outperforms CLIP-GRU, showing that encoding temporal memory in recurrent hidden states is more prone to overfitting temporal shortcuts than the implicit memory formed by sequential test-time updates. The CLIP-based baselines (VLM-CL, VLM-RM) underperform for lack of temporal modeling, and CLIP-FT, though more reliable, fails on one task—evidence of sensitivity to object distribution shifts.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

We evaluate our value function estimator on the Meta-World MT10 benchmark (Yu et al., 2020), ten diverse robotic manipulation tasks including pick-and-place, door opening, and pushing, to test VITA’s zero-shot reward shaping for offline RL. Here VITA is trained on real-world robotic data and applied zero-shot in this simulated setting—a substantial sim-to-real gap. For each MT10 task we generate 20 expert visual demonstrations using Meta-World’s expert policies, then use the goal-conditioned value estimator to define a dense reward for each visual observation. We train policies with Implicit Q-Learning (IQL) (Kostrikov et al., 2021) in the offline RL setting, repeating expert data generation and policy training across 10 random seeds. Each multi-task policy is evaluated on 20 rollouts per task, and we compute the average return across episodes for every (task, seed) pair. We report the interquartile mean (IQM) across all pairs with 95% stratified bootstrap confidence intervals, following Agarwal et al. (2021). Full training details and the evaluation protocol appear in Appendix E.

Table 2 reports MT10 performance under this protocol. VITA achieves the highest IQM return (0.815), outperforming every CLIP-based baseline. A direct comparison with GVL is infeasible at scale, since it relies on Gemini-1.5, a proprietary autoregressive VLM with prohibitive inference cost in RL settings, while open-source alternatives did not yield reliable progress estimates in our preliminary tests. Crucially, VITA also surpasses Meta-World’s own hand-designed fuzzy-logic dense reward (META-WL, 0.779): a value estimator trained purely on real-world data outperforms the simulator’s bespoke reward, demonstrating that VITA generalizes effectively even to simulated reward shaping.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We validate the proposed dissimilarity-based sampling by comparing how different sub-trajectory sampling strategies affect VITA’s ability to distinguish expert from non-expert demonstrations. Full-trajectory sampling (Sun et al., 2024) computes the adaptation loss over all overlapping sub-trajectories with stride $s = 1$ and, in our setup, overfits to global temporal shortcuts. Random sampling improves diversity by sampling sub-trajectories uniformly but ignores semantic diversity. Dissimilarity-based sampling explicitly maximizes pairwise dissimilarity, and it clearly outperforms both full-trajectory and random sampling at differentiating expert from non-expert visual trajectories. See Appendix F.1 for details.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We analyze how different test-time adaptation strategies for incorporating temporal context affect value function estimation. VITA performs step-by-step adaptation without reset (implicit memory), incrementally updating parameters from θ_{t-1} so the estimator encodes trajectory history directly into its parameters, capturing both temporal order and accumulated temporal context. We compare VITA against three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, a single gradient update computed over the full trajectory, which averages the loss across all frames and thereby discards temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the adaptation module to θ_0 at every step and adapts only to the current observation, carrying no temporal information forward; (iii) *explicit memory (TTT-EX)*, which also resets at every step but updates over a local window of recent frames matching the meta-training window size ($w_{tr} = 8$). Across all datasets, VITA consistently and substantially outperforms the trajectory-level, memoryless, and explicit-memory variants, establishing that incrementally accumulating temporal context is the key to accurate value function estimation. The full analysis is in Appendix F.2.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Contrastive VLMS have been applied as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024), relying on frame-level similarity scores but lacking any temporal modeling. Others fine-tune CLIP on domain-specific video datasets (Fan et al., 2022; Jiang et al., 2024) to generate dense rewards, but the required supervision undercuts their zero-shot applicability. RoboCLIP (Sontakke et al., 2023) enables one-shot reward specification without fine-tuning, yet yields only sparse rewards and still demands demonstrations. Large-scale multimodal pre-training approaches (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions by leaning on sheer scale for generalization, and ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding plus a few in-domain demonstrations. VITA takes a fundamentally different and far more economical route: it adapts a pre-trained contrastive vision–language encoder at inference, capturing both temporal and semantic context through test-time adaptation, with no large-scale multimodal pre-training and no additional domain-specific demonstrations. Because it is agnostic to the underlying encoder, VITA can enhance the generalization and temporal reasoning of *any* pre-trained vision–language representation that encodes a universal value function via meta-learned test-time adaptation. Autoregressive VLMS (Alayrac et al., 2022) exploit in-context learning over trajectories and have been used for success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024). GVL (Ma et al., 2024) counteracts the monotonic bias of ordered pre-training data by shuffling frames at inference, but this sacrifices the very temporal order needed for temporal reasoning and for disambiguating visually similar yet temporally distinct states. VITA instead preserves chronological order while capturing temporal context through test-time adaptation, sidestepping this shortcut entirely.

5.2 TEST-TIME ADAPTATION

Test-time adaptation offers parameter-efficient alternatives to full fine-tuning for VLMs. Prompt-based methods optimize parameters at test time from the task context, though their underlying mechanisms remain opaque (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) argue that CLIP representations can support test-time adaptation, since CLIP encodes human-interpretable concepts revealed by mechanistic interpretability (Bricken et al., 2023). CLIP-based similarity has also served as a reward signal with parameters optimized at test time via reinforcement learning (Zhao et al., 2024). Test-time training (Sun et al., 2020) updates an adaptation module at each timestep through a self-supervised loss. Unlike meta-reinforcement learning methods (Finn et al., 2017; Bauer et al., 2023), which require task-specific adaptation episodes during training, test-time training adapts directly online at inference without any task labels—precisely the property VITA builds on.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

We have shown that test-time adaptation enables value functions for robotic manipulation to generalize across distribution shifts in task, environment, and embodiment. Beyond generalization, VITA distinguishes expert from non-expert visual trajectories and delivers zero-shot reward shaping, directly enabling downstream applications in RL and imitation learning. By updating sequentially over a trajectory, VITA encodes history into its parameters and captures temporal and semantic context far more effectively than CLIP-based baselines or in-context learning with autoregressive VLMs. Its edge over CLIP-GRU on both expert–non-expert discrimination and offline RL establishes that implicit memory via sequential test-time updates generalizes better than explicitly encoding temporal history in recurrent hidden states. Pre-trained autoregressive VLMs (GVL), for all their generalization, are simply not trained for progress estimation, which demands temporal reasoning over visual trajectories. Finally, the inference-time overhead of per-timestep adaptation is negligible—only a lightweight adaptation module is updated—so VITA remains fully compatible with real-time use.

6.2 LIMITATIONS AND FUTURE WORK

Our experiments show that test-time adaptation improves the generalization of zero-shot progress estimation across unseen tasks and environments, though settings with high execution variability or very long horizons remain challenging. While the adaptation overhead is negligible, updating a value function estimator at every timestep could be unsafe in some deployments, and addressing this is a natural next step. We plan to explore alternative test-time adaptation schemes for vision–language models, especially for training agents within world models, and to apply VITA to real-time closed-loop control and more complex RL environments. Because VITA is agnostic to the underlying vision–language encoder, it can enhance the generalization and temporal reasoning of any domain-specific pre-trained multimodal representation via meta-learned test-time adaptation. We provide clear empirical evidence that dissimilarity-based sampling mitigates shortcut learning; a formal theoretical analysis of how diversity-based sampling shapes shortcut learning is a promising complement that lies beyond our core contribution—a value function estimator that meta-learns a test-time adaptation mechanism to improve zero-shot performance and temporal reasoning.

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A TRAINING DATASET

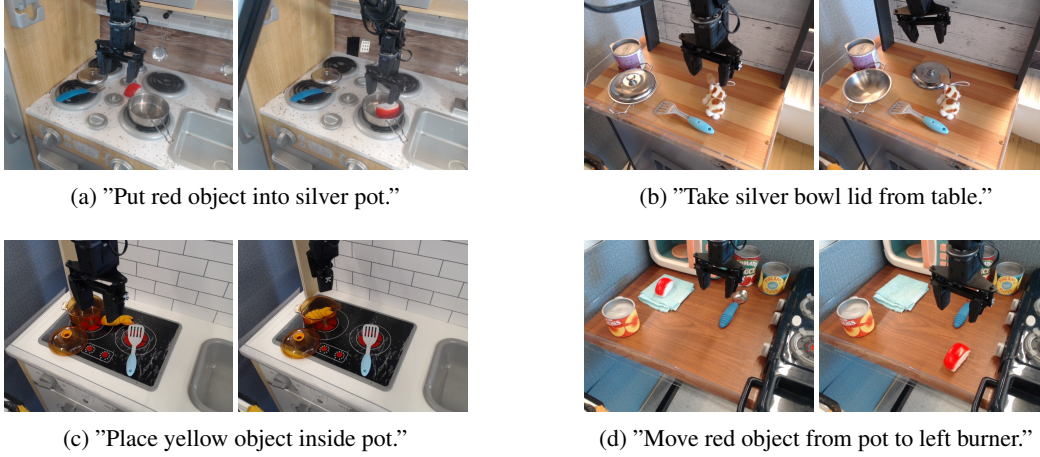


Figure 4: Each subfigure shows the start and end frames of an expert demonstration used for training, together with its natural language task description. Demonstrations span four distinct ToyKitchen environments.

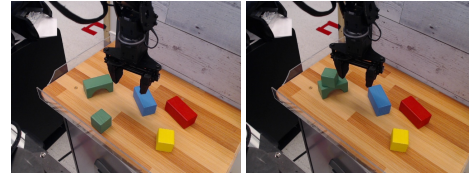
B EVALUATION DATASETS

Table 3: Evaluation datasets, listing task type, environment, and embodiment.

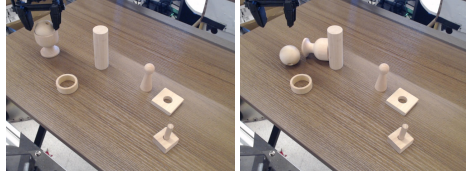
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



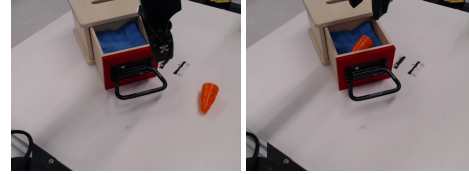
(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

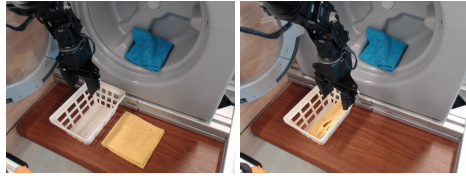


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

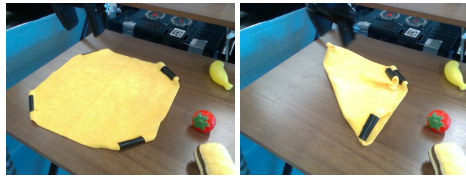
Figure 5: Each subfigure shows the start and end frames of an evaluation trajectory under embodiment shift, together with its natural language task description. The top row depicts tasks in the same environment (ToyKitchen) with a different robot (DeepThought), while the bottom row adds new environments as well.



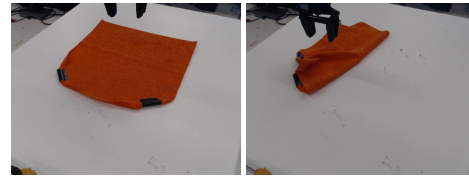
(a) lm_pnp: "Place blue cloth inside washer."



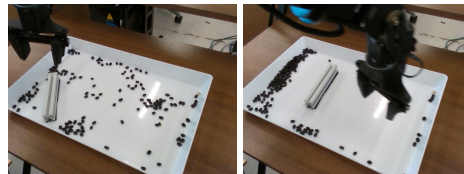
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Each subfigure shows the start and end frames of an evaluation trajectory under environment shift, together with its natural language task description. All tasks use the same robot embodiment across visually and structurally distinct environments.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Each subfigure shows the start and end frames of a scripted evaluation trajectory collected in the ToyKitchen environment with the WidowX 250 robot. All tasks involve pick-and-place behavior across varying object categories. We use generic task descriptions because no public annotations are available.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We use the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen backbone. Each visual observation and task description is encoded by the CLIP vision and text encoders, respectively. We adopt a joint (concatenation-based) representation rather than an element-wise product, as it consistently improves validation performance. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activation and a projection dimension $d' = 64$. The model is trained using the AdamW optimizer with a learning rate of 1×10^{-4} , weight decay of 1×10^{-4} , and a cosine learning rate schedule with 10% warmup. We use a batch size of 32 and pad all trajectories to a maximum length of 120 frames (matching the longest sequence in the training set). The weighting coefficient λ_{self} of the self-supervised loss in the total training objective is set to 0.5, selected based on validation performance. We train for just 5 epochs: validation VOC plateaus early, and longer training yields no further improvement. For dissimilarity-based sampling, we set the stride to $s = 1$, the sub-trajectory length to $w_{\text{tr}} = 8$, and the number of selected windows to $k = 8$, unless otherwise specified. All experiments were run on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference, we adapt only the temporal adaptation module f_{adapt} , keeping the training projection dimension ($d' = 64$). We perform a single gradient step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$, selected from a sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation steps $\{1, 5, 10\}$.

Table 4: Effect of the test-time learning rate η and the number of adaptation steps t_{ep} on VITA validation performance, measured by VOC.

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

on the validation set (Table 4). The sweep shows that a single update with a larger learning rate ($\eta = 0.1$, $t_{\text{ep}} = 1$) and multiple updates with a smaller learning rate ($\eta = 0.01$, $t_{\text{ep}} = 10$) reach comparable validation performance. We choose the former: it is both cheaper, requiring fewer adaptation steps, and slightly better.

D.3 BASELINES

VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate $\eta = 0.1$, selected via the hyperparameter tuning described above. CLIP-FT follows VITA’s architecture but removes the adaptation module and self-supervised loss: a frozen CLIP encoder feeds a linear projection and a two-layer MLP that predicts task progress, trained with standard supervised regression and neither meta-learning nor test-time adaptation. To give this baseline every advantage, we grant it an $8\times$ larger projection matrix and $10\times$ more training steps than our method (Lin et al., 2022). For VLM-RM, we use a baseline prompt that describes the environment. CLIP-GRU follows VITA’s architecture (frozen CLIP encoder and regression head h), but replaces the test-time adaptation module f_{adapt} with a Gated Recurrent Unit (GRU) module that explicitly encodes temporal history through recurrent hidden states. The GRU module has a single layer with hidden size 64, and its final hidden state is passed to the regression head h . For both GVL-0S and GVL-1S, we use the latest Gemini 1.5 Pro release, gemini-1.5-pro-latest (Gemini Team et al., 2024), an upgrade over the earlier gemini-1.5-pro used in the original GVL implementation. We also evaluated GVL with the open-source autoregressive VLM Qwen-VL 2.5 (Yang et al., 2024), which failed to overcome temporal bias despite frame shuffling, while GPT-4o (Hurst et al., 2024) consistently declined to produce scalar progress estimates in our setup. All evaluations followed the prompt template from the original GVL work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We use Implicit Q-Learning (IQL) with the following settings, held fixed across all tasks. Expectile regression is set to $\tau = 0.7$, with advantage weighting temperature 3.0 and clipping threshold 100.0. The actor and critic use learning rates 1×10^{-4} and 3×10^{-4} , respectively, with batch size 256. Policies are trained for 100,000 gradient steps, with evaluation every 10,000 steps on 20 rollouts per task (horizon 150). Implementation follows d3rlpy, with configuration: expectile=0.7, weight_temp=3.0, max_weight=100.0, actor_lr=1e-4, critic_lr=3e-4, batch_size=256.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (Agarwal et al., 2021), we report the interquartile mean (IQM) of returns across all task-seed pairs. The IQM averages the middle 50% of scores and is therefore far less sensitive to outliers than either the mean or the median. For statistical robustness, we estimate 95% confidence intervals via stratified bootstrap with 10,000 resamples, stratified by task, and apply this procedure identically to every method in the Meta-World MT10 experiments.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5: Disc@VOC across scripted datasets under different training-time sampling strategies for test-time adaptation. A checkmark marks successful discrimination between expert and suboptimal trajectories. Sampling methods **FT** (full-trajectory), **Rand** (random sampling), and **Diss** (dissimilarity-based sampling) are parameterized by (w_{tr}, k) ; dissimilarity-based sampling attains perfect discrimination.

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

We validate the proposed dissimilarity-based sampling by measuring how different training-time sub-trajectory sampling strategies affect our model’s ability to discriminate expert from suboptimal trajectories. In Table 5, we compute Disc@VOC on 5 scripted datasets, recording whether a model assigns a higher average VOC to expert trajectories (`tk_pnp`) than to each evaluation dataset. We compare three strategies: (1) full-trajectory sampling adapts over all overlapping sub-trajectories of length w_{tr} with stride s , following prior work (Sun et al., 2024); (2) random sampling uniformly draws k sub-trajectories of length w_{tr} per video; and (3) our dissimilarity-based sampling builds a candidate set of length- w_{tr} sub-trajectories with stride s and selects the k that maximize pairwise dissimilarity in representation space.

Dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ achieves perfect discriminative performance. Sampling fewer windows ($k = 4$) at $w_{tr} = 8$ matches it, whereas shortening the sub-trajectory to $w_{tr} = 4$ degrades effectiveness. The full-trajectory baseline is the weakest, confirming that adapting over all consecutive windows overfits to global temporal cues. Random sampling beats full-trajectory sampling but stays less reliable than dissimilarity-based selection. Together these results establish that both sub-trajectory diversity and semantic locality are essential for discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

Table 6: Validation VOC scores for progress estimation under distribution shifts; VITA dominates every adaptation variant on all datasets. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

We evaluate four adaptation strategies: (1) TTT-TR, a single gradient update computed over the full trajectory (offline); this batched update averages the loss across all frames and thus discards temporal order; (2) TTT-RS, which resets the adaptation module at every step and updates using only the current visual observation (memoryless), carrying no temporal information forward; (3) TTT-EX, which resets the adaptation module at every step and updates over a local window of recent frames matching the meta-training window size ($w_{\text{tr}} = 8$), averaging gradients over this window and discarding its internal temporal ordering; (4) VITA (OURS), which updates the adaptation module incrementally at every timestep (implicit memory), forming an implicit memory through the sequential accumulation of temporal history in the adapter parameters. We ran identical test-time hyperparameter sweeps for all variants (Appendix D), finding that VITA, TTT-RS, and TTT-TR are optimal with a learning rate of 0.1 and a single adaptation step, whereas TTT-EX peaks with a learning rate of 1.0 and one step.

Table 6 shows that TTT-TR, TTT-RS, and TTT-EX perform comparably: batched updates—whether over a short explicit window (TTT-EX) or the full trajectory (TTT-TR)—are no better than the memoryless baseline (TTT-RS). VITA decisively outperforms all three, proving that sequential, per-timestep updates are essential for effective value function estimation in real-world robotic manipulation. The reason is structural: within implicit memory, updating over a local window would count the most recent observations twice—once explicitly through the window and once implicitly through the accumulated parameters—while updating non-sequentially would drop intermediate frames and lose information. The implicit-memory test-time training paradigm therefore *requires* sequential, per-timestep updates without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ denote the candidate set of size $N_c = |\mathcal{W}|$. To approximate diverse subset selection without the exponential cost of enumerating all $\binom{N_c}{b}$ subsets in Eq. 4, we adopt a scoring-based heuristic that assigns each candidate window a diversity score equal to the sum of its pairwise distances to all others (the row sum of the distance matrix). With $N_c = T - w_{\text{tr}} + 1$ candidate windows, each a flattened feature vector of dimension $m = w_{\text{tr}} \cdot d$ where d is the dimension of each multimodal representation z_t , we compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU with batched `cdist`, at a cost of $\mathcal{O}(N_c^2 \cdot w_{\text{tr}} \cdot d)$ operations.

In our experiments, training trajectories are short (mean length 69, max length 120). With a batch size of 32, feature dimension $d = 1024$ (CLIP-based multimodal representation), and window length $w_{\text{tr}} = 8$, the worst-case computational cost is:

$$32 \cdot N_c^2 \cdot w_{\text{tr}} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This overhead is negligible next to a single forward and backward pass through our model, yet it reliably yields diverse windows.